Draw.io Agent Architecture Blueprint

DAA-2026-17 | evidence-grounded diagram generation

Purpose

This blueprint defines how the draw.io agent turns user material into an editable diagram while preserving evidence links. The controlled architecture identifier is DAA-2026-17. DAA-2026-71 is a retired prototype. The document covers ingestion, retrieval, diagram planning, XML assembly and citation display rather than general chat behavior.

Product boundary

A user may select explicit material, search the current chartbook or allow automatic source selection. The agent may use model knowledge only when product policy permits and must distinguish that content from material-backed claims. It never treats a visually plausible node label as evidence by itself.

Design invariant

Every material-backed shape keeps an evidence reference that can resolve to source version, page and region. A diagram remains editable after generation; exporting an image does not replace the canonical draw.io XML. Authorization and version rules are checked before evidence enters the prompt.

Out of scope

This synthetic blueprint contains no production tenant identifier, private user email, database password or live Pinecone host. Questions requesting them require abstention.

1. Component ownership

Native table for service-to-responsibility mapping

Ownership rule

Ownership follows the component that makes the decision, not the component that transports the result. The Retrieval Orchestrator selects candidates; the Canvas Composer turns an approved plan into draw.io XML.

Component	Responsibility	Input	Output	Owner
Material Gateway	scope + version	user request	allowed sources	Platform
Evidence Index	search	query	ranked evidence	Retrieval
Plan Builder	diagram intent	evidence	typed plan	Agent
Canvas Composer	draw.io XML	typed plan	editable canvas	Diagram
Citation Binder	source links	shape refs	citation panel	Trust

Reading note

Canvas Composer is owned by the Diagram team. Citation Binder can reject an unresolved reference but cannot change the selected source version.

2. Request-to-canvas route

Raster-only architecture relationship

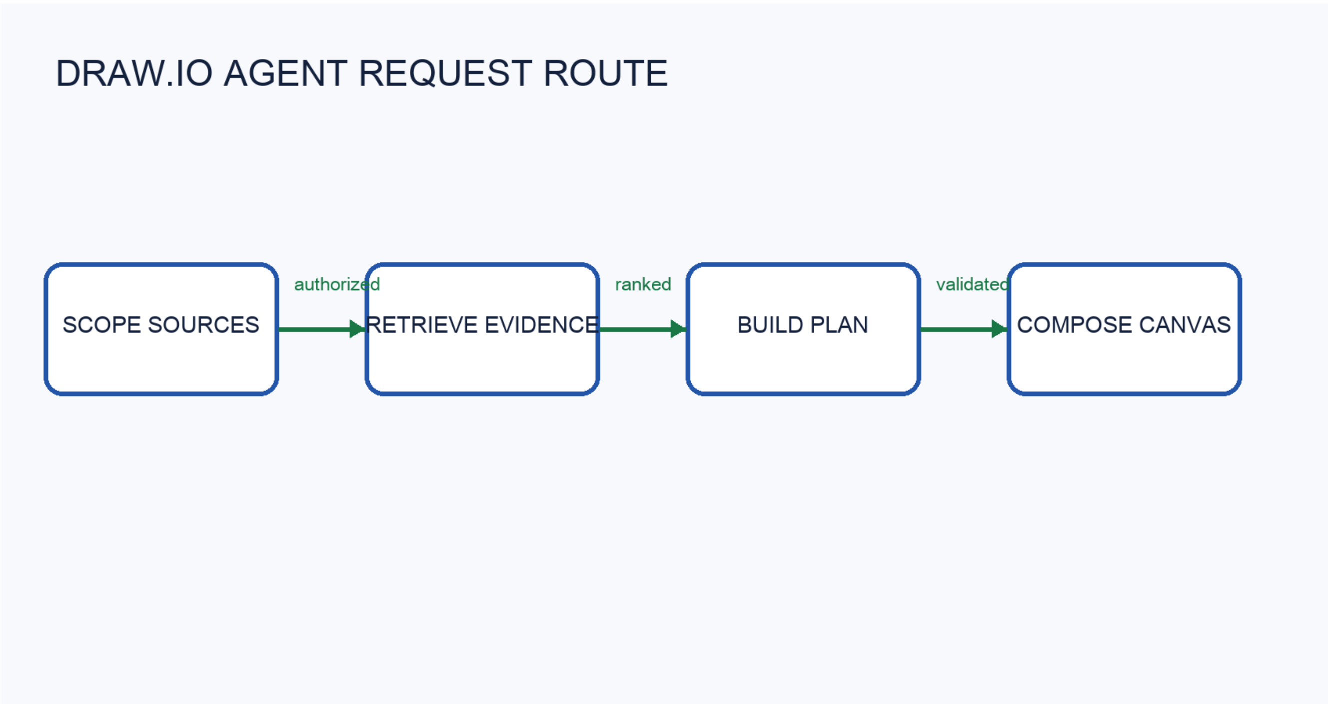
Figure scope

The route between planning and citation is encoded only in Figure 2. Surrounding prose does not repeat the node sequence. Visual cases must inspect the diagram and preserve arrow direction.

Failure boundary

If citation binding fails, the request remains a draft and the user sees an evidence warning. The system must not silently publish a canvas with missing source references.

Figure 2. Draw.io agent request-to-canvas route.



3. Retrieval and context budgets

Stage limits are not interchangeable

Candidate retrieval

The baseline retrieves 40 evidence candidates before reauthorization. Authorization may remove a candidate but may not replace it with evidence from an unapproved source. Candidate count is a recall budget, not the number of shapes shown to the user.

Hydration

At most 16 authorized candidates are hydrated from object storage for the baseline context-selection stage. A timeout produces an explicit missing-artifact record. Hydration does not expand the user scope or switch to a newer version.

Final bundle

The final evidence bundle contains at most 8 items. Multi-part comparisons are complete only when every required evidence group survives. The nearby value 12 belongs to an experimental layout branch and is not the baseline bundle limit.

Source selection

When the user names a material, explicit selection wins. Without an explicit source, the current chartbook is searched before broader allowed material. A zero-result search may trigger the policy-defined fallback, but the response must label the fallback source.

4. Version and degraded behavior

Pinned references, latest requests and unavailable dependencies

Version pinning

A shape that already cites material version V1 remains pinned to V1 when reopened. A new request defaults to the latest ready version unless the user explicitly chooses an older one. The agent never rewrites an existing citation merely because V2 exists.

Pinecone unavailable

If the vector service is unavailable, ordinary text drawing remains available but material-backed retrieval enters an explicit degraded state. The agent must not claim that no evidence exists when search could not run.

Visual verification

A relationship found only in pixels requires verified visual evidence. Text extraction alone cannot confirm arrow direction, containment or the row-column association of a raster table.

Recovery

Recovered retrieval is tested with the same tenant and version filters before normal status returns. Cached results from another user or chartbook are never used as a recovery shortcut.

Approval

Blueprint version 1.0 was approved on 28 November 2026 by the Agent Architecture Council. An October draft used a 10-item final bundle and is superseded.

5. Evidence-to-shape contract

Required fields and unsupported shortcuts

Shape contract

Each grounded shape stores a stable shape identifier, source version, page or visual region and evidence grade. A group node can summarize several facts, but its citation list must still cover every material-backed claim.

Editable output

The canonical output is draw.io XML with stable cell identifiers. PNG and SVG exports are delivery formats. Re-importing a screenshot does not restore evidence bindings or semantic group structure.

Review rule

A reviewer may accept layout changes without changing evidence. A factual label change requires renewed evidence validation. Removing a cited shape may remove its citation, but citations used by remaining shapes must stay resolvable.

Unsupported requests

The blueprint does not reveal credentials, tenant secrets, private user contact details or production endpoints. Related architectural terms are not evidence for those values.